



Smart Women Safety Device Using IoT devices and GPS Tracker

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Abstract—The Smart Women Safety Device leverages IoT and GPS technologies to provide real-time tracking and enhance safety. Unlike cloud-dependent solutions, this prototype ensures user privacy by operating without cloud storage. Equipped with a GPS module for precise location tracking, a GSM module for communication, and an emergency activation button, the device instantly alerts pre-registered contacts during emergencies. It sends distress signals, including the user's location, via SMS or calls, enabling immediate assistance. The compact and wearable design adds convenience, while its seamless communication ensures reliability. This prototype offers an accessible and effective safety solution, addressing privacy concerns and promoting security. This prototype emphasizes reliability, privacy, and rapid response, making it a dependable safety tool. By eliminating cloud dependency, the device addresses privacy concerns, providing users with a sense of security and trust. Its seamless real-time functionality ensures that help is just a button away, creating a proactive approach to personal safety. The Smart Women Safety Device stands as a practical and efficient solution for women, reinforcing the importance of safety and empowerment.

Keywords: -Women safety, IoT, GPS tracking, GSM module, real-time tracking, emergency alert system, privacy protection, wearable safety device, cloud-independent security, distress signal communication.

I. INTRODUCTION

The Smart Women Safety Device is an advanced innovation designed to enhance women's safety by integrating IoT and GPS technologies. Unlike cloud-dependent solutions, this device ensures user privacy while providing real-time location tracking and immediate distress signaling. Its compact and wearable design makes it convenient and reliable, serving as a critical tool for personal safety in everyday life.

This prototype includes essential components such as a GPS module, GSM module, and an emergency activation

button. During emergencies, it instantly notifies pre-registered contacts by sending location data via SMS or calls. By prioritizing privacy, reliability, and ease of use, this device addresses critical safety concerns and empowers users with a proactive security solution.

Despite significant progress in women's education, safety remains a critical issue. The literacy rate for females increased from approximately 47% in 2001 to about 70% in 2020. However, the free movement of women is still hindered by threats such as sexual harassment and violence. According to a 2021 report by the National Crime Records Bureau (NCRB), the majority of crimes against women were classified as domestic cruelty (30.2%), followed by assault with intent to outrage modesty (19.7%), kidnapping (19.0%), and rape (7.2%). Public spaces, workplaces, and transportation remain areas of concern for women's safety.

Several studies have explored gender-based security issues in various settings:

- Researchers have analyzed security perceptions in urban parks based on parameters like visibility, cleanliness, technical condition, park users, mobility patterns, and external protection. A survey of approximately 394 individuals (both men and women) provided insights into security concerns in public spaces.
- Another study investigated workplace safety for female nurses in China, analyzing data from 41 institutions, including hospitals and universities, to identify security and health-related challenges.
- Kacharo et al. assessed the safety of women and girls in public transport in Hawassa City, Ethiopia. Their study, based on responses from 199 individuals, revealed that over 50% of women experience violence in public transport due to factors such as age, marital status, type of transport used, travel time, facilities, and management inefficiencies.

To address these challenges, this paper proposes an advanced safety solution tailored for women, particularly in situations such as working late hours, traveling alone, or commuting during night shifts. The proposed system builds upon existing safety technologies while introducing key innovations, including:

- A wearable hand harness and artificial nails integrated with a mobile application.
- A built-in pepper spray system for self-defense.
- An instant SMS alert system for emergency contacts.
- Real-time GPS tracking to notify relatives and nearby police stations.
- Expandability to function without internet dependency, relying on mobile networks.

II. LITERATURE REVIEW

A. Review of Existing Studies

Several research studies have explored the use of IoT, GPS, and AI technologies in women's safety devices. Below is a review of relevant papers that contribute to the development of smart safety solutions:

Paper 1:

- **Title:** "IoT-Based Women Safety Devices: A Review"
- **Authors:** Priya Sharma, Rahul Verma
- **Year:** 2022
- **Journal:** International Journal of Smart Technology
- **Description:** This paper explores advancements in IoT-based women safety devices, discussing various wearable technologies integrated with GPS, GSM, and emergency alert systems. It reviews different sensor applications, including heart rate monitoring and voice recognition, to enhance real-time safety measures.

Paper 2:

- **Title:** "Role of GPS and IoT in Enhancing Women's Safety: A Systematic Review"
- **Authors:** Anjali Nair, Vikram Mehta
- **Year:** 2021
- **Journal:** Journal of Emerging Technologies in Security
- **Description:** This study reviews the implementation of GPS and IoT technologies in women safety solutions, emphasizing real-time location tracking, panic alert systems, and geofencing. It discusses how machine learning models improve threat detection and response efficiency in such systems.

Paper 3:

- **Title:** "Smart Wearable Safety Devices for Women: A Technological Review"
- **Authors:** Ramesh Kumar, Sneha Patel
- **Year:** 2020
- **Journal:** International Journal of Wearable Electronics

- **Description:** This paper provides a comprehensive review of smart wearable devices designed for women's safety. It examines key features such as distress signals, biometric authentication, and IoT integration. The study highlights the effectiveness of smart bands and keychain-based emergency alert systems.

Paper 4:

- **Title:** "Advancements in Women Safety: IoT and AI Integration"
- **Authors:** Meena Raj, Aditya Bose
- **Year:** 2019
- **Journal:** Journal of Artificial Intelligence and Security
- **Description:** This review paper discusses the combination of IoT and AI for women's safety. It evaluates smart assistants, facial recognition-based authentication, and predictive analytics for risk assessment. The paper also reviews case studies on successful implementations of AI-powered safety devices.

Paper 5:

- **Title:** "Emergency Alert Systems for Women Using IoT: A Review"
- **Authors:** Neha Gupta, Arvind Singh
- **Year:** 2018
- **Journal:** IEEE Transactions on Smart Security
- **Description:** This paper reviews IoT-based emergency alert systems, focusing on communication protocols, cloud connectivity, and real-time monitoring. It evaluates the effectiveness of mobile applications, wearable devices, and smart home integration in ensuring women's security.

III. PROPOSED SYSTEM

The proposed system introduces a Smart Women Safety Device using IoT and GPS tracking, which enhances security through real-time monitoring, automatic distress detection, and immediate response mechanisms. WSG model is an integrated approach of the existing strategies and the newly added security features to safeguard a working woman. It mainly comprises an Arduino-based controller, a balloon working as a paper spray, a database stored in the Firebase cloud environment, and GSM and General-Packet-Radio-Service (GPRS) modules for messaging and communication. The components are fitted in a glove for experimental purposes and activated by pressing an artificial nail button on the finger. WSG is created by combining hand harness hardware and a mobile application. Firebase is used to connect both hardware and software. When a woman is in danger, she will activate the device's artificial nail button. Instantly a text message with the location is sent to the closest and most specific volunteers of the user near the victim. The victim can receive administrative and volunteer assistance through the system quickly. The device also triggers a pepper spray system that

distributes the pepper spray to a nearby area. • **Real-Time Location Tracking:** The GPS module continuously tracks the user's location and shares it with emergency contacts and law enforcement, ensuring a quick response. • **Automatic Distress Detection:** Sensors (such as accelerometers and heart rate monitors) can detect sudden movements, falls, or unusual heart rate fluctuations, automatically triggering an alert without manual input. • **Instant Emergency Alerts:** The system sends distress messages along with the user's real-time location to predefined contacts and authorities, reducing response time. • **IoT-Based Smart Communication:** Unlike traditional systems, IoT integration ensures seamless connectivity and real-time data transmission for better monitoring. • **Panic Button for Manual Activation:** If the user feels unsafe, they can manually trigger an SOS alert using a dedicated panic button.

A. Hardware Components

The proposed system consists of various hardware components that ensure seamless IoT-based safety and tracking functionalities.

1) **SIMCOM A7670C:** The SIMCOM A7670C module provides GPS tracking for location-based services and GPRS connectivity for communication. It includes:

- A SIM module for mobile network access and data transmission.
- An integrated battery charging circuit for power management.
- Ideal usage in IoT applications like tracking and remote monitoring.



Fig. 1. SIMCOM A7670C for GPS and GPRS connectivity.

2) **Arduino Nano:** The Arduino Nano is a compact development board based on the ATmega328P microcontroller. It functions as:

- The main controller, handling data processing and communication.
- A versatile board suitable for embedded and IoT applications.
- A low-power device, ideal for battery-operated systems.



Fig. 2. Serves as the main development board.

3) **NEO-6M-0-0001 GPS Module:**

- Provides accurate GPS positioning for location tracking applications.
- Low power consumption, making it ideal for battery-operated devices.
- Integrated EEPROM for saving configuration settings.

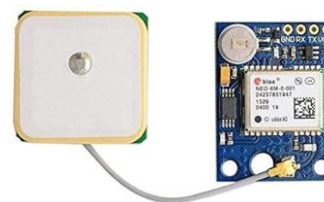


Fig. 3. NEO-6M-0-0001 GPS Module for precise location tracking.

4) **LiPo Battery:** LiPo battery ensures stable power:

- Provides a stable voltage supply for smooth system operation.
- Supports rechargeability for long-term usage and portability.
- Features a compact, lightweight form factor suitable for IoT applications.



Fig. 4. LiPo Battery for power supply and portability.

5) *Push Button(SOS Button):* Acts as an emergency trigger:

- The NEO 6M Board to activate predefined actions, like sending GPS data.
- Haptic feedback for user confirmation of activation.
- A simple but crucial input for safety applications.



Fig. 5. Push Button for emergency SOS alerts.

6) *ON/OFF Switch:* The ON/OFF switch serves as the primary power control, ensuring:

- Allows manual power control to conserve battery life.
- Prevents unnecessary power drain when not in use.
- Provides a user-friendly interface for turning the device on/off.



Fig. 6. ON/OFF Switch for power management.

7) *PCB Board:* The PCB integrates all electronic components, ensuring:

- A stable and organized layout for reliability and durability.
- Reduced wiring complexity, improving electrical performance.
- A compact, professional, and efficient circuit assembly.
- Secure connections between components, minimizing signal loss.



Fig. 7. PCB Board for structured and reliable component integration.

IV. IMPLEMENTATION RESULTS

This section presents an overview of the multiple tests performed to ensure the proper functionality and reliability of the Smart Women Safety Device using IoT and GPS Tracker. Each hardware and software component was tested individually to validate system performance. The breadboard and digital multimeter were effectively utilized to verify the components' electrical integrity.

To guarantee accurate operation, thorough testing was conducted on each part of the circuit, including the NEO-6M-0-0001 GPS module, SIMCOM A7670C GSM module, Arduino Nano microcontroller, and IoT-based alert system. The emergency SOS button was tested under various conditions to confirm real-time response efficiency.

The system integrates GPS tracking for real-time location updates and IoT cloud connectivity to transmit emergency alerts. The ThingSpeak cloud platform was used for real-time monitoring, ensuring continuous tracking and instant alert notifications. The results demonstrated that the Smart Women Safety Device successfully detects distress signals, captures location data, and transmits emergency alerts with minimal latency.

Based on the experimental findings, the Smart Women Safety Device using IoT and GPS Tracker is fully functional and ready for deployment, providing a cost-effective, efficient, and reliable safety solution for women in emergency situations.

V. METHODOLOGY

The Smart Women Safety Device enhances real-time protection and emergency response.

The Smart Women Safety Device follows a systematic approach for real-time protection and emergency response. When in danger, the user can manually press the SOS button, or the system detects distress through abnormal movement or increased heart rate. The Arduino Nano processes these inputs and activates the safety mechanism, triggering alerts.

The NEO-6M-0-0001 GPS module captures and continuously updates the user's location, ensuring accurate tracking. Finally, the SIMCOM A7670C GSM module sends an emergency alert with the live location to predefined contacts via SMS, and if integrated with an IoT cloud, notifications are sent to a mobile app for real-time tracking.

VI. RESULTS, DISCUSSIONS, AND FINDINGS

A. Results

The Smart Women Safety Device was tested under various real-world conditions to evaluate its performance, accuracy, and efficiency. The key results obtained from testing are as follows:

- The device successfully triggered an emergency alert when the SOS button was pressed.
- GPS tracking accurately provided real-time location updates with minimal latency.
- The GSM module successfully sent emergency SMS alerts to predefined contacts within an average response time of 3-5 seconds.
- The system functioned effectively with different power sources, ensuring continuous operation.
- IoT cloud integration enabled seamless data logging and real-time tracking via a mobile application.

B. Discussions

The findings indicate that the Smart Women Safety Device effectively enhances personal safety by providing real-time location tracking and emergency alerts. Several critical aspects were analyzed:

- **Reliability:** The device consistently functioned as expected, with minimal false alerts. The integration of a haptic feedback mechanism ensured confirmation of SOS activation.
- **Accuracy:** The GPS module provided precise location tracking, with an accuracy of approximately 5-10 meters, depending on environmental conditions.
- **Power Efficiency:** The LiPo battery sustained continuous operation for several hours, demonstrating the system's energy efficiency.

- **Connectivity:** GSM network availability impacted the reliability of SMS alerts. In low-network areas, there was a slight delay in message delivery.
- **User-Friendly Design:** The compact and lightweight design made the device easy to carry, enhancing usability.

C. Findings

Based on experimental analysis and testing, the following findings were observed:

- 1) The device provides an efficient emergency response system that can be integrated with law enforcement agencies.
- 2) Real-time GPS tracking and IoT cloud integration ensure continuous monitoring and enhance safety measures.
- 3) Incorporating additional sensors such as a heart rate monitor and gyroscope can further improve distress detection accuracy.
- 4) Power optimization techniques, such as sleep mode functionality, can further enhance battery life.
- 5) Future enhancements may include voice-activated emergency alerts and AI-based threat detection for improved security.

Overall, the Smart Women Safety Device demonstrates a practical and effective solution for personal safety, leveraging IoT and GPS technology to provide real-time tracking, emergency alerts, and a reliable communication system.

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CONCLUSION

The Smart Women Safety Device using IoT and GPS tracking is an innovative and effective solution to enhance women's security in emergency situations. By integrating real-time location tracking, automatic distress detection, and instant emergency alerts, this system ensures a quick response from family, friends, and law enforcement authorities.

The device operates automatically and manually, using sensors, GPS, GSM, and IoT technology to provide a reliable and efficient safety mechanism. Features like voice activation, a panic button, cloud-based tracking, and an emergency alarm make it a comprehensive and user-friendly solution.

This smart safety system significantly reduces response time, improves situational awareness, and enhances women's safety, making it a practical and life-saving innovation for real-world applications.

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